

AI, Automation and Open Web Workflows

Converting scattered knowledge into reusable workflows, diagrams, and publishable web material.

Rajesh Kumar

Assistant Professor, Department of Physics
Sido Kanhu Murmu University, Dumka, Jharkhand
M.Sc. Physics, IIT Kanpur
kr.rajesh.phy@gmail.com

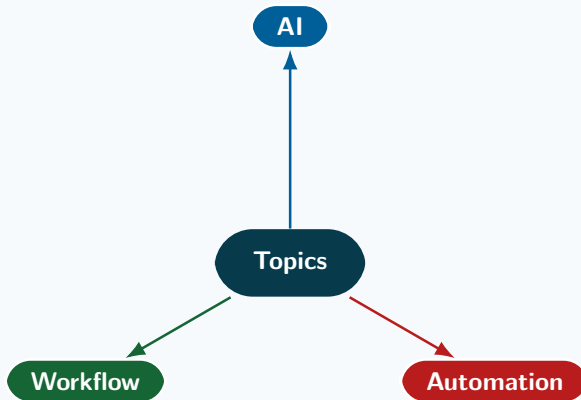
UGC MMTTC Refresher Course

June 2026

Refresher Course in Information Technology



Topics We Will Discuss



1

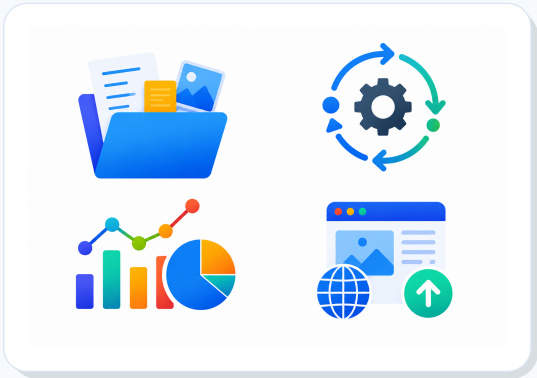
Artificial Intelligence

What comes to mind when you think of AI?



AI USE Framework

- ▶ **Organize:** Collect scattered notes, PDFs, screenshots, and ideas into clear folders, chapters, slides, or web posts.
- ▶ **Automate:** Make repeated work one-command, such as downloading PDFs, merging files, fetching weather, or compiling LaTeX.
- ▶ **Visualize:** Turn ideas into diagrams, charts, Beamer slides, TikZ figures, and clean academic templates.
- ▶ **Publish:** Convert organized material into PDFs, websites, GitHub pages, blogs, and reusable teaching resources.



Repetitive Tasks; Connecting Tools

AI can generate code

- ▶ AI can write code in many programming languages.
- ▶ Latex, Python, HTML, JavaScript, and more.
- ▶ AI can also explain code and debug errors.



I bought a computer,
but **unknowingly accepted**
the **OS choice** made for me.

**SAME COMPUTER MARKET.
DIFFERENT AWARENESS.**

I chose the computer for its quality,
then installed a **free OS**
of my choice.

I want a good quality computer, but **WITHOUT OS.**

**COMPUTER
WITHOUT OS
(NO LICENCE COST)**

VS

**PRE-INSTALLED
WINDOWS
READY TO USE**

Windows 11



Not Aware



OS comes pre-installed



Pay More



Less Freedom



OPERATING SYSTEM IS ALSO A CHOICE.
AWARENESS TURNS A BUYER INTO A DECISION-MAKER.



Freedom



Security



Save Money



Community



Control

AI BRIDGES LINUX EXPERTISE AND WINDOWS AUTOMATION

Your Command-Line Skills. **AI Translates.** Windows Executes.



LINUX EXPERTISE

Strong command-line thinking, shell logic, and automation skill.

```
user@linux:~$ ls -l
total 48
drwxr-xr-x 2 user user 4096 May 12 10:11 docs
drwxr-xr-x 2 user user 4096 May 12 10:12 scripts
-rw-r--r-- 1 user user 20485760 May 12 10:15 lecture.mp4
-rw-r--r-- 1 user user 2048576 May 12 10:15 notes.txt
user@linux:~$
```

LINUX
Command
Logic

```
ffmpeg -i lecture.mp4 -vn lecture.mp3
```

```
find . -name "*.pdf"
```

```
grep grep -R "result"
```

```
for f in *.jpg; do mv "$f" "img_$f"; done
```

```
rsync -av folder/ backup/
```

In Linux, you think in commands and get things done.

AI BRIDGE

ChatGPT • Gemini • Claude



WINDOWS
Automation
Execution

- Translate command (Bash → PowerShell / CMD)
- Adapt syntax for Windows environments
- Explain errors and suggest fixes
- Convert Bash scripts to PowerShell / Batch
- Generate scripts and one-liners
- Suggest Windows alternatives
- Debug commands and outputs
- Automate workflows end-to-end



WINDOWS AUTOMATION

AI helps apply Linux expertise inside Windows workflows.

```
PS C:\Works> Get-ChildItem

Mode                LastWriteTime         Length Name
----                -
-a-----  5/12/2024  18:15       10485760  lecture.mp4
-a-----  5/12/2024  18:18       2048576   notes.txt
-a-----  5/12/2024  18:20              0  script.ps1
PS C:\Works>
```

```
ffmpeg -i lecture.mp4 -vf scale=1280:720 output.mp4

Get-ChildItem *.jpg | Rename-Item -NewName
{ "photo_" + $_.Name }

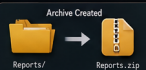
Get-ChildItem -Recurse -Filter *.pdf
Select-String -Path *.txt -Pattern "result"
```

```
Ccobocopy D:\Work E:\Backup /MIR
```

```
Compress-Archive -Path Reports -DestinationPath Reports.zip
```

In Windows, you execute powerful automation with confidence.

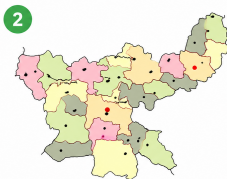
RESULTS: REAL WORK, AUTOMATED



Linux and Windows are different, but **AI reduces the command-line gap.**

AI helps you translate knowledge, adapt commands, and automate across both worlds.

Coding For Weather Forecast Report Download



```
# Download each page as a PDF
for id in ${IDS[@]}; do
  BROWSER --headless --disable-gpu \
  --print-to-pdf="${id}.pdf" \
  "${URL_BASE}?id=${id}"
done
```

```
IDS=(id1 id2 id3 ... idN)
BROWSER=chrome | chromium | google-chrome
URL_BASE=https://city.ind.gov.in/...
```



```
# Crop each PDF
for id in ${IDS[@]}; do
  pdfcrop --margins "TOP RIGHT BOTTOM LEFT" \
  "${id}.pdf" "cropped_${id}.pdf"
done
```

TOP RIGHT BOTTOM LEFT → margin values (points)



```
# Combine the cropped PDFs
pdftitle FIRST_PDF \
$(for id in ${IDS[@]}; do echo -n "cropped_${id}.pdf "; done) \
OUTPUT_PDF
```

FIRST_PDF → cover page (e.g., A.pdf)

OUTPUT_PDF → final combined file (e.g., combined.pdf)



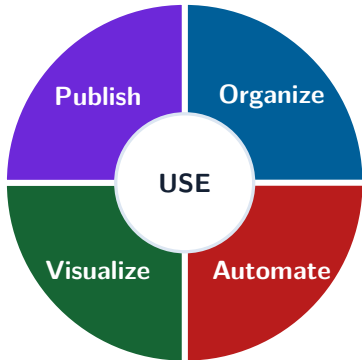
```
# Clean up & save final PDF
find . -type f -name '*.pdf' \
| -name 'OUTPUT_PDF' | -name 'FIRST_PDF' -exec rm {} +
cp OUTPUT_PDF TARGET_DIR/Weather-$(date +%F).pdf
cd TARGET_DIR
open Weather-$(date +%F).pdf
```

OUTPUT_PDF → combined file name (e.g., combined.pdf)

FIRST_PDF → cover page (e.g., A.pdf)

TARGET_DIR → destination directory

Workflow Easily Achieved Using Coding Agents



Codex



Claude

2

Automation

Task Types for Automation

1



2



3



5



1. Downloading Monthly Journals

2. Opening News

3. Image and Video Files Processing

4. Accessing Files in One Click

Reduce Image Size using FFmpeg

ffmpeg**-i "\$1"****Input Image File****\$1**

(Original image path)

-vf "scale=iw/2:-1"**Scale Width to Half****iw/2:-1**

(Half width, height auto)

-q:v "\$2"**Set Output Quality****\$2**

(Quality value 1-31)

r_"\$1"**Output File****r_"\$1"**

(New file with prefix r_)

HOW IT WORKS – STEPS

1**Provide input image file**
Specify the path of the image ("1").**2****Reduce width to 50%**
scale=iw/2:-1 makes width half,
height adjusts automatically.**3****Set output quality**
-q:v "2" sets compression quality.
1 = best quality, 31 = lowest quality.**4****Save new image**
Output is saved with prefix r_"1"
to avoid overwriting original.

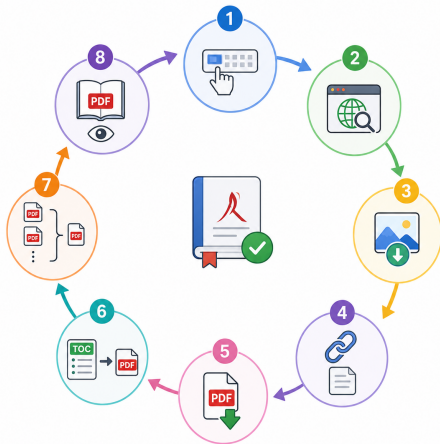
BEFORE → AFTER

Original Image**4000 × 3000**
~ 2.8 MB**Reduced Image****2000 × 1500**
~ 250 KB (Example)**EXAMPLE USAGE**

```
ffmpeg -i "photo.jpg" -vf "scale=iw/2:-1" -q:v 8 r_"photo.jpg"
```

Input: photo.jpg
Quality: 8 (High)
Output: r_photo.jpg (smaller size)

Automating Journal Download



Journal issue to book-form PDF

1. Enter volume and issue number.
2. Check whether the PDF already exists.
3. Fetch the page numbers from the website.
4. Extract article page ranges and titles.
5. Download the cover page of the issue.
6. Build direct PDF links for all articles.
7. Download all article PDFs.
8. Create a table-of-contents.
9. Merge cover, TOC, and articles in one PDF.
10. Remove temp files and open the final PDF.

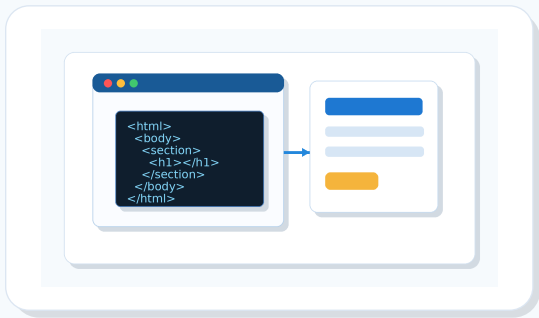
3

Web Workflows

What Is a Web Page?

Web page = HTML structure

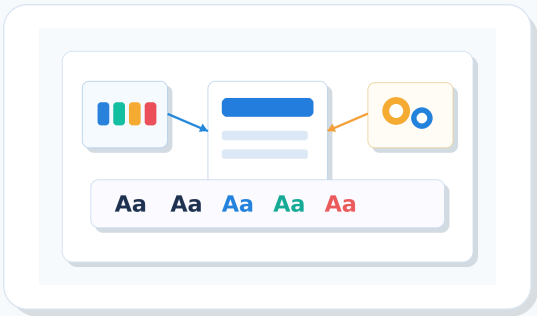
- ▶ A web page is built from HTML.
- ▶ HTML gives the page its basic structure.
- ▶ Headings, paragraphs, links, images, and sections are HTML elements.
- ▶ What we see in the browser has a code background.



Styles, Functions and Fonts

CSS and JavaScript add life

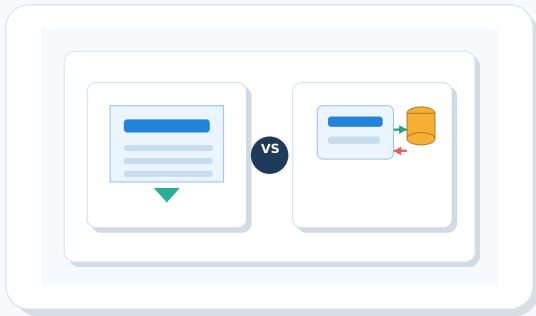
- ▶ CSS controls colors, spacing, fonts, cards, and layout.
- ▶ JavaScript controls interaction and dynamic behavior.
- ▶ Bootstrap gives ready-made design components.
- ▶ Together they turn plain HTML into a modern page.



Static vs Dynamic Websites

Two kinds of websites

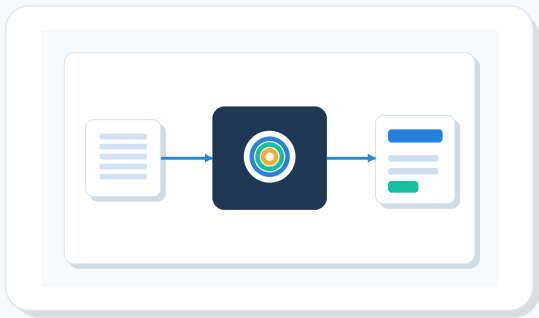
- ▶ Static websites are pre-built pages served directly.
- ▶ Dynamic websites generate pages using databases and server logic.
- ▶ Static sites are faster, simpler, and easier to host.
- ▶ Dynamic sites are useful when content changes for every user.



Jekyll as a Workflow Solver

Write once, organize automatically

- ▶ Jekyll converts Markdown into web pages.
- ▶ Posts, pages, categories, tags, and links become organized.
- ▶ GitHub Pages can publish the site directly.
- ▶ This makes scattered notes reusable and publishable.



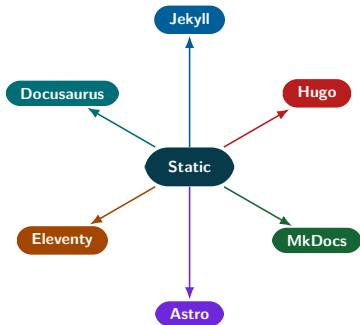
4

Publishing with Jekyll and GitHub Pages

Why Jekyll?

Jekyll solves the publishing problem

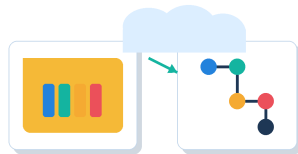
- ▶ Write content once in Markdown.
- ▶ Jekyll converts it into a static website.
- ▶ Posts, pages, categories, and layouts become organized automatically.
- ▶ Other popular static-site tools include Hugo, MkDocs, Astro, Eleventy, and Docusaurus.



GitHub as a Free Repository

GitHub stores your project files

- ▶ A repository is a project folder stored online.
- ▶ It can keep code, notes, images, PDFs, and website files.
- ▶ GitHub supports version history and collaboration.
- ▶ Public repositories can be used freely for open academic material.



GitHub Pages: Free Website Hosting

From repository to website

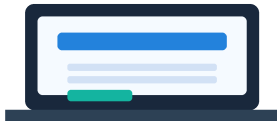
- ▶ GitHub Pages publishes static websites from a repository.
- ▶ It has built-in support for Jekyll.
- ▶ A user website can appear as `username.github.io`.
- ▶ This is useful for teaching notes, academic pages, blogs, and documentation.



My Website

Type this URL in your browser

rajeshphy.github.io



rajeshphy.github.io

Thank You

Questions and Discussion

AI, Automation and Open Web Workflows

Rajesh Kumar

Assistant Professor, Department of Physics
Sido Kanhu Murmu University, Dumka, Jharkhand

rajeshphy.github.io

kr.rajesh.phy@gmail.com